Chapter 12 solutions

CH12-E1

1.
$$\overline{x} = \frac{3+5+5+9}{4} = \frac{22}{4} = 5.5.$$

2. The deviations are $-2.5, -0.5, -0.5, 3.5$, whose squared sum is
$$6.25 + 0.25 + 0.25 + 12.25 = 19.$$

Therefore
$$s^2 = \frac{19}{4-1} = \frac{19}{3}.$$

3. With denominator $N = 4$,
$$\frac{19}{4} = 4.75.$$

4. The $N-1$ statistic is unbiased under i.i.d. sampling with finite variance and $N \geq 2$. Division by N describes the observed set. For i.i.d. Gaussian observations with both mean and variance unknown, joint maximum likelihood also uses the sample mean and denominator N. If the mean is known, center at that known mean instead.

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CH12-E2

1.
$$\mathbb{E}[X] = 0.2(0) + 0.5(1) + 0.3(2) = 1.1.$$

2.
$$\mathbb{E}[X^2] = 0.2(0) + 0.5(1) + 0.3(4) = 1.7.$$
$$\mathrm{Var}(X) = 1.7 - (1.1)^2 = 1.7 - 1.21 = 0.49.$$

3.
$$\mathbb{E}[Y] = 3(1.1) - 2 = 1.3,$$
$$\mathrm{Var}(Y) = 9(0.49) = 4.41.$$

4. The calculation in part 2 directly checks
$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

CH12-E3

1. $X \sim U(0, L)$:
$$f(x) = \begin{cases} 1/L, & 0 \leq x \leq L, \\ 0, & \text{otherwise}, \end{cases}$$

Chapter 12 solutions (continued)

$$F(x) = \begin{cases} 0, & x < 0, \\ x/L, & 0 \le x \le L, \\ 1, & x > L. \end{cases}$$

2. $L = 4$:

$$P(1 < X < 3) = F(3) - F(1) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

3.

$$\mathbb{E}[X] = L/2 = 2 \quad (L = 4), \qquad \mathrm{Var}(X) = \frac{L^2}{12} = \frac{4}{3} \quad (L = 4).$$

4. Translation gives $Y = a + X$, so

$$\mathbb{E}[Y] = a + \mathbb{E}[X], \qquad \mathrm{Var}(Y) = \mathrm{Var}(X).$$

The interval width is unchanged, as is the variance.

CH12-E4

1. Row sums:

$$P(X = 0) = 0.3, \qquad P(X = 1) = 0.7.$$

Column sums:

$$P(Y = 0) = 0.5, \qquad P(Y = 1) = 0.5.$$

2.

$$\mathbb{E}[X] = 0.7, \qquad \mathbb{E}[Y] = 0.5.$$

Only the cell $(1, 1)$ contributes to the product expectation.

$$\mathbb{E}[XY] = 0.4.$$

3.

$$\mathrm{Cov}(X, Y) = 0.4 - (0.7)(0.5) = 0.05.$$

$$\mathrm{Var}(X) = 0.7 - 0.7^2 = 0.21,$$

$$\mathrm{Var}(Y) = 0.5 - 0.5^2 = 0.25.$$

$$\rho_{XY} = \frac{0.05}{\sqrt{0.21}\sqrt{0.25}} \approx 0.2182.$$

4. Independence would require

$$P(X = 0)P(Y = 0) = 0.3 \cdot 0.5 = 0.15.$$

The actual joint mass is 0.2, proving dependence.

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Chapter 12 solutions (continued)

CH12-E5

1. Three equally probable columns:

$$\mu = \frac{1}{3}\left[\begin{bmatrix}1\\1\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix} + \begin{bmatrix}-1\\0\end{bmatrix}\right] = \begin{bmatrix}0\\2/3\end{bmatrix}.$$

2. Centered columns:

$$z_1 = (1, 1/3)^T, \qquad z_2 = (0, 1/3)^T, \qquad z_3 = (-1, -2/3)^T.$$

3.

$$V = \frac{1}{3}(z_1 z_1^T + z_2 z_2^T + z_3 z_3^T).$$

Outer-product sums:

$$\sum_k (z_k)_1^2 = 1 + 0 + 1 = 2,$$

$$\sum_k (z_k)_1 (z_k)_2 = \frac{1}{3} + 0 + \frac{2}{3} = 1,$$

$$\sum_k (z_k)_2^2 = \frac{1}{9} + \frac{1}{9} + \frac{4}{9} = \frac{2}{3}.$$

Therefore

$$V = \begin{bmatrix}2/3 & 1/3\\1/3 & 2/9\end{bmatrix}.$$

4.

$$\det V = \frac{2}{3}\cdot\frac{2}{9} - \frac{1}{9} = \frac{1}{27} > 0.$$

The first leading diagonal entry and determinant are positive, proving positive definiteness. The centered possible values are not collinear, so no nonzero coefficient vector annihilates all their projections.

CH12-E6

1. $\mu = (1, 2)^T$, $\det V = 4$:

$$p(x) = \frac{1}{4\pi}\exp\left[-\frac{1}{2}\left(\frac{(x_1 - 1)^2}{4} + (x_2 - 2)^2\right)\right].$$

2. Level curve:

$$\frac{(x_1 - 1)^2}{4} + (x_2 - 2)^2 = \text{constant}.$$

The first semiaxis is twice the second because the variances are 4 and 1.

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Chapter 12 solutions (continued)

3. $A = \mathrm{diag}(1, 3)$:

$$\mathbb{E}[Y] = A\mu = (1, 6)^T,$$
$$V_Y = AVA^T = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}.$$

4. For example, choose

$$V' = \begin{bmatrix} 4 & 3 \\ 3 & 9 \end{bmatrix}.$$

This has correlation $1/2$ and determinant 27, with positive leading entry, so it is SPD.

$$p(x) = \frac{1}{2\pi\sqrt{27}} \exp\left[-\frac{1}{2}(x-\mu)^T (V')^{-1}(x-\mu)\right].$$

CH12-E7

1.

$$J(x) = \frac{(b_1 - x)^2}{\sigma_1^2} + \frac{(b_2 - x)^2}{\sigma_2^2}.$$

2.

$$J'(x) = -2\frac{b_1 - x}{\sigma_1^2} - 2\frac{b_2 - x}{\sigma_2^2} = 0.$$

Hence

$$\widehat{x} = \frac{b_1/\sigma_1^2 + b_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2}.$$

3.

$$\mathrm{Var}(\widehat{x}) = \frac{1}{1/\sigma_1^2 + 1/\sigma_2^2} = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}.$$

4. The variances are 1 and $9/4$, giving precision weights 1 and $4/9$.

$$\widehat{x} = \frac{10 + (4/9)11}{1 + 4/9} = \frac{134}{13} \approx 10.3077.$$

CH12-E8

1. $V^{-1} = \mathrm{diag}(1, 1/4, 1/9)$, whitened matrix:

$$\tilde{A} = V^{-1/2}A = \begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \\ 0 & 1/3 \end{bmatrix}.$$
$$A^T V^{-1} A = \tilde{A}^T \tilde{A} = \begin{bmatrix} 5/4 & 1/4 \\ 1/4 & 13/36 \end{bmatrix}.$$

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Chapter 12 solutions (continued)

The weighted equations are

$$\begin{bmatrix} 5/4 & 1/4 \\ 1/4 & 13/36 \end{bmatrix} x = A^T V^{-1} b.$$

2. Determinant:

$$\Delta = \frac{5}{4} \cdot \frac{13}{36} - \left(\frac{1}{4}\right)^2 = \frac{56}{144} = \frac{7}{18}.$$

$$(A^T V^{-1} A)^{-1} = \frac{18}{7} \begin{bmatrix} 13/36 & -1/4 \\ -1/4 & 5/4 \end{bmatrix} = \begin{bmatrix} 13/14 & -9/14 \\ -9/14 & 45/14 \end{bmatrix}.$$

3. BLUE requires a fixed full-column-rank design, a correct linear mean, zero-mean errors, and a known SPD covariance structure, possibly up to a common scale. Gaussianity is not required. If only a common scale is unknown, the estimator is unchanged but its covariance retains that scale.
4. Options include a stated shrinkage model, regularization, dimension reduction, pooled covariance, or a suitable robust covariance estimator. These do not all guarantee invertibility automatically. Report the estimator, tuning parameter, effective rank, sample size, and inference consequences.

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CH12-E9

1. The centered matrix has shape $M \times n$, so

$$\operatorname{rank}(X_c X_c^T) \le \min(M, n).$$

In fact $X_c \mathbf{1} = 0$ forces rank at most $\min(M, n-1)$, and the $n-1$ covariance estimator requires $n \ge 2$. If $M \ge n$, it is necessarily singular. Using it directly as a Gaussian covariance gives no ordinary full-dimensional Gaussian density; additional modeling is required.

2. Linearity gives mean μ. Independence with finite second moments removes both cross-covariance terms, so

$$\operatorname{Cov}(X + Z) = V + W.$$

If X is also Gaussian, the independent sum is $N(\mu, V + W)$. General non-Gaussian data plus Gaussian noise need not become exactly Gaussian, although the stated moments still hold.

3. If

$$X_2 = aX_1 + b,$$

then

$$\operatorname{Cov}(X_1, X_2) = a \operatorname{Var}(X_1),$$

Chapter 12 solutions (continued)

$$\operatorname{Var}(X_2) = a^2 \operatorname{Var}(X_1).$$

With positive finite $\operatorname{Var}(X_1)$ and $a \neq 0$, correlation is $+1$ for positive a and -1 for negative a, and covariance rank is 1. If $a = 0$, the second variable is constant and correlation is undefined. If both are constant, covariance rank is zero.

4. Covariance describes joint linear variation, not intervention effects. A common-cause illustration is hot weather increasing both ice-cream sales and swimming exposure, potentially making sales and drowning counts move together without a causal effect of purchasing ice cream. A regression omitting such a common cause need not have a causal coefficient.

CH12-E10

1. $V = V^T$, with leading minors $3 > 0$, $\det V = 9 - 1 = 8 > 0$, proving positive definiteness. The characteristic polynomial is

$$\det(V - \lambda I) = \lambda^2 - 6\lambda + 8,$$

Hence $\lambda_+ = 4, \lambda_- = 2$. For $\lambda = 4$,

$$(V - 4I)z = 0 \iff -z_1 + z_2 = 0,$$

take $q_+ = (1,1)^T/\sqrt{2}$. For $\lambda = 2$,

$$z_1 + z_2 = 0,$$

take $q_- = (1,-1)^T/\sqrt{2}$. They are orthogonal unit vectors.

Since $V = Q\,\mathrm{diag}(4,2)Q^T$, whitening

$$W = V^{-1/2}E = Q\,\mathrm{diag}(1/2, 1/\sqrt{2})Q^T E$$

scales q_+ by $1/2$ and q_- by $1/\sqrt{2}$, producing covariance I.

2. First compute

$$A^T V_e^{-1} A = \begin{bmatrix} 5/4 & 1/4 \\ 1/4 & 13/36 \end{bmatrix},$$
$$A^T V_e^{-1} b = A^T (2, 5/4, 4/9)^T = (13/4, 61/36)^T.$$

The weighted normal equations are

$$\begin{bmatrix} 5/4 & 1/4 \\ 1/4 & 13/36 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 13/4 \\ 61/36 \end{bmatrix}.$$

Their coefficient inverse is

$$(A^T V_e^{-1} A)^{-1} = \begin{bmatrix} 13/14 & -9/14 \\ -9/14 & 45/14 \end{bmatrix},$$

Hence

Chapter 12 solutions (continued)

$$\widehat{x} = \left(\frac{27}{14}, \frac{47}{14}\right)^T \approx (1.928571, 3.357143)^T.$$

The prediction is

$$A\widehat{x} = \left(\frac{27}{14}, \frac{74}{14}, \frac{47}{14}\right)^T = \left(\frac{27}{14}, \frac{37}{7}, \frac{47}{14}\right)^T,$$

The residual is

$$e = b - A\widehat{x} = \left(\frac{1}{14}, -\frac{2}{7}, \frac{9}{14}\right)^T.$$

Whiten componentwise by standard deviations.

$$\tilde{e} = V_e^{-1/2} e = \left(\frac{1}{14}, -\frac{1}{7}, \frac{3}{14}\right)^T.$$

Also $V_e^{-1} e = (1, -1, 1)^T/14$, so $A^T V_e^{-1} e = (0, 0)^T$, checking weighted orthogonality. The objective is $\|\tilde{e}\|_2^2$.

$$J = \frac{(1/14)^2}{1} + \frac{(-2/7)^2}{4} + \frac{(9/14)^2}{9} = \frac{1}{14}.$$

3. OLS uses $A^T A x = A^T b$.

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad A^T b = (7, 9)^T.$$

Hence

$$x_{\text{OLS}} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 7 \\ 9 \end{bmatrix} = \left(\frac{5}{3}, \frac{11}{3}\right)^T.$$

Its difference from WLS is

$$x_{\text{OLS}} - \widehat{x} = \left(\frac{5}{3} - \frac{27}{14}, \frac{11}{3} - \frac{47}{14}\right) = \left(-\frac{11}{42}, \frac{13}{42}\right) \approx (-0.261905, 0.309524).$$

WLS gives the high-variance third measurement less weight, producing different estimates.

4. Sufficient BLUE assumptions are the correct model $b = Ax + e$, fixed full-column-rank A, $\mathbb{E}[e] = 0$, and the stated $V_e = \text{diag}(1, 4, 9) \succ 0$. Whitening, weighted normal equations, unbiasedness, and the Gauss–Markov covariance ordering do not require normality. For covariance $\sigma^2 V_e$ with unknown $\sigma^2 > 0$, the estimate is unchanged and its covariance is multiplied by σ^2. Jointly normal errors additionally permit exact normal sampling distributions and likelihood inference. Estimating general weights from the same data makes them random, so the fixed-weight proof no longer applies directly.

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Chapter 12 solutions (continued)

CH12-E11

1. $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 2/3$, giving $\mathrm{Var}(X) = 2/3$. Also $\mathbb{E}[Y] = 2/3$, $\mathbb{E}[Y^2] = \mathbb{E}[X^4] = 2/3$, so $\mathrm{Var}(Y) = 2/9$. Symmetry gives $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0$ and zero covariance. Yet $P(Y = 0 \mid X = 0) = 1 \neq P(Y = 0) = 1/3$, proving dependence. Both variances are positive, so correlation is defined and equals zero.
2. $S_N \sim \mathrm{Bin}(N, 1/2)$, with mean $N/2$ and variance $N/4$. Thus $z_k = (2k - N)/\sqrt{N}$, spacing $\Delta z = 2/\sqrt{N}$, and probabilities $\binom{N}{k}2^{-N}$. Bars of width Δz and height $\binom{N}{k}2^{-N}/\Delta z$ have the correct probability areas, totaling 1 by the binomial theorem. Density heights likewise have probability per coordinate length. The finite distribution remains discrete.
3. Integration gives

$$\mathbb{E}[X^2] = \frac{1}{2}\int_{-1}^{1} x^2\,dx = \frac{1}{3}, \quad \mathbb{E}[X^4] = \frac{1}{2}\int_{-1}^{1} x^4\,dx = \frac{1}{5}.$$

Thus $\mathrm{Var}(X^2) = 1/5 - 1/9 = 4/45$. Independence gives estimator mean $1/3$ and variance $4/(45N)$. Chebyshev gives

$$P(|\widehat{\mu} - 1/3| \geq 0.01) \leq \frac{4}{45N(0.01)^2}.$$

To make the bound at most 0.05, it suffices that $N \geq 4/(45 \cdot 10^{-4} \cdot 0.05) = 17777.\overline{7}$, so the smallest sufficient integer from this bound is 17778. This is not claimed to be necessary for the actual distribution.

4. No. Distributional convergence makes CDF error vanish at each fixed continuity point of the limit. It gives no common $N = 30$ guarantee for all finite-variance distributions. Quantitative Berry–Esseen bounds require more information, such as a finite absolute third moment. Rare extreme observations can delay a good normal approximation even when variance is finite.

CH12-E12

1. Unbiasedness and uncorrelated new and old errors make the joint covariance block diagonal when the old estimate is treated as a summary measurement with covariance W_0. Write its normal equation as $W_0^{-1}\widehat{x}_0 = g_0$. Adding new data gives coefficient $W_1^{-1} = W_0^{-1} + H^T R^{-1} H$ and right-hand side $g_0 + H^T R^{-1} b_1$. Subtract $W_1^{-1}\widehat{x}_0$.

$$W_1^{-1}(\widehat{x}_1 - \widehat{x}_0) = H^T R^{-1}(b_1 - H\widehat{x}_0).$$

Chapter 12 solutions (continued)

Since $W_1^{-1} \succ 0$, multiply by W_1 to obtain the update with $K = W_1 H^T R^{-1}$. For $S = R + HW_0H^T \succ 0$, directly check

$$W_1^{-1} W_0 H^T S^{-1} = H^T R^{-1}(R + HW_0H^T)S^{-1} = H^T R^{-1},$$

proving $K = W_0 H^T S^{-1}$.

2. First $S = 10, K = 1/10$, so $\widehat{x}_1 = 10 + (1/10)(13 - 10) = 103/10$, $W_1 = 1 - 1/10 = 9/10$. Next $S = 19/10, K = 9/19$, giving

$$\widehat{x}_2 = \frac{103}{10} + \frac{9}{19}\left(9 - \frac{103}{10}\right) = \frac{184}{19}, \quad W_2 = \frac{9}{10}\left(1 - \frac{9}{19}\right) = \frac{9}{19}.$$

The batch result is $(10 + 13/9 + 9)/(1 + 1/9 + 1) = 184/19$, with covariance $(1 + 1/9 + 1)^{-1} = 9/19$, exactly matching.

3. Since $W_0 - W_1 = W_0 H^T S^{-1} H W_0$, its quadratic form at any z is $(HW_0z)^T S^{-1}(HW_0z) \geq 0$, proving $W_1 \preceq W_0$. A copied observation has correlated error and nonzero off-diagonal covariance blocks, so the block-diagonal update is invalid. Copying one noisy measurement adds no information; assigning variance $R/2$ to two copies falsely assumes independence.

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Chapter 12 solutions (continued)

Verification code appendix: statistics_checks.py

```
"""Recompute sample moments, covariance and weighted least-squares examples."""

from math import exp, isclose, pi, sqrt
from fractions import Fraction


def mean(values: tuple[float, ...]) -> float:
    return sum(values) / len(values)


def sample_variance(values: tuple[float, ...]) -> float:
    average = mean(values)
    return sum((value - average) ** 2 for value in values) / (len(values) - 1)


def inverse2(matrix: tuple[tuple[float, ...], ...]) -> tuple[tuple[float, ...], ...]:
    """Use exact rational arithmetic on stored finite values, then convert to float; not a
general numerical solver."""
    p, q, r, s = (Fraction(matrix[i][j]) for i, j in ((0, 0), (0, 1), (1, 0), (1, 1)))
    determinant = p * s - q * r
    if determinant == 0:
        raise ValueError("exactly singular matrix for the supplied values")
    return (
        (float(s / determinant), float(-q / determinant)),
        (float(-r / determinant), float(p / determinant)),
    )
```

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Chapter 12 solutions (continued)

```
assert inverse2(((1e-8, 0.0), (0.0, 1e-8))) == ((1e8, 0.0), (0.0, 1e8))
try:
    inverse2(((1.0, 2.0), (2.0, 4.0)))
except ValueError:
    pass
else:
    raise AssertionError("a truly singular matrix must be rejected")


def matvec(matrix: tuple[tuple[float, ...], ...], vector: tuple[float, ...]) -> tuple[float,
...]:
    return tuple(
        sum(matrix[i][j] * vector[j] for j in range(len(vector)))
        for i in range(len(matrix))
    )


samples = (3.0, 5.0, 5.0, 9.0)
assert mean(samples) == 5.5
assert isclose(sample_variance(samples), 19.0 / 3.0)

joint = ((0.2, 0.1), (0.3, 0.4))
px = tuple(sum(row) for row in joint)
py = tuple(sum(joint[i][j] for i in range(2)) for j in range(2))
assert all(isclose(px[i], value, abs_tol=1e-12) for i, value in enumerate((0.3, 0.7)))
assert all(isclose(py[i], value, abs_tol=1e-12) for i, value in enumerate((0.5, 0.5)))
ex = px[1]
```

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Chapter 12 solutions (continued)

```
ey = py[1]
exy = joint[1][1]
covariance = exy - ex * ey
assert isclose(covariance, 0.05)

states = ((1.0, 1.0), (0.0, 1.0), (-1.0, 0.0))
state_mean = tuple(sum(state[i] for state in states) / 3 for i in range(2))
state_covariance = tuple(
    tuple(sum((state[i] - state_mean[i]) * (state[j] - state_mean[j]) for state in states) /
3 for j in range(2))
    for i in range(2)
)
assert isclose(state_covariance[0][0], 2 / 3)
assert isclose(state_covariance[0][1], 1 / 3)
assert isclose(state_covariance[1][1], 2 / 9)
assert isclose(state_covariance[0][0] * state_covariance[1][1] - state_covariance[0][1] ** 2,
1 / 27)

covariance_matrix = ((4.0, 0.0), (0.0, 9.0))
covariance_inverse = inverse2(covariance_matrix)
for triangle in ((1.0, 0.0), (0.0, 1.0), (2.0, 3.0)):
    mahalanobis = triangle[0] * matvec(covariance_inverse, triangle)[0] + triangle[1] *
matvec(
        covariance_inverse, triangle
    )[1]
    assert mahalanobis > 0

sigma1 = 1.0
sigma2 = 1.5
b1 = 10.0
```

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Chapter 12 solutions (continued)

```
b2 = 11.0
weights = (1.0 / sigma1**2, 1.0 / sigma2**2)
estimate = (weights[0] * b1 + weights[1] * b2) / sum(weights)
assert isclose(estimate, 134.0 / 13.0)
estimate_variance = 1.0 / sum(weights)
assert isclose(estimate_variance, 9.0 / 13.0)

# Check that even weighting is a special case.
ordinary_sample_variance = 1.0 / (2.0 / (sigma1**2 + sigma2**2))
assert ordinary_sample_variance > 0

# Numerically integrate the standard one-dimensional Gaussian on a bounded grid.
step = 0.01
grid_sum = (
    sum(exp(-0.5 * (k * step) ** 2) for k in range(-1000, 1001))
    * step
    / sqrt(2.0 * pi)
)
assert isclose(grid_sum, 1.0, rel_tol=2e-3)

print("sample variance =", sample_variance(samples))
print("joint covariance =", covariance)
print("weighted estimate =", estimate)
print("weighted estimate variance =", estimate_variance)
```

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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